

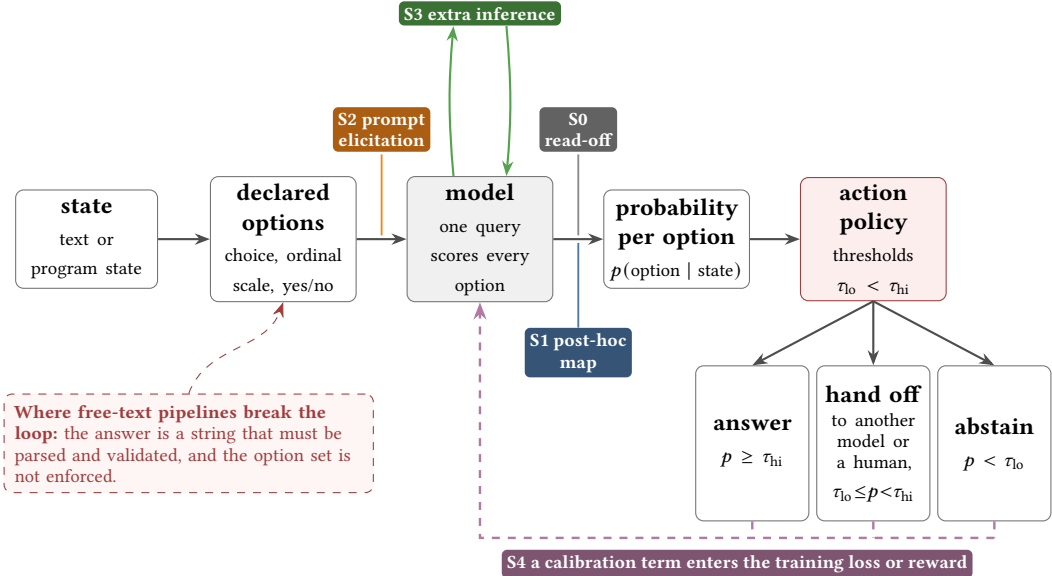


Fig. 10. **The calibrated decision loop, and where each stage injects calibration.** A decision model receives a state and a *declared* option set, scores every option in one query, and hands a probability per option to an explicit action policy. The policy drawn here answers when the probability is high, hands a less certain case off to another model or a human, and abstains when the probability is low; Table 2 lists the other low-confidence actions (trigger retrieval, filter the output, adapt compute, return a prediction set). The five stages of the taxonomy differ in where calibration enters: the model’s own read-off (**S0**), a post-hoc map (**S1**), a prompt that asks for confidence (**S2**), extra inference such as sampling or a conformal set (**S3**), or a calibration term in the model’s own training loss or reward (**S4**). The dashed callout marks where free-text generation pipelines break the loop (Table 2). Hosted and open typed-decision models (Tables A1 and A2) expose this interface: a state, declared options (a choice among labels, an ordinal scale or yes/no) and a probability per option.